

Mathematical Explanation of mcmc2.py

The script is a Bayesian inversion. It searches for fault parameters that explain observed InSAR line-of-sight displacement data.

1. Unknown Model Vector

Each MCMC sample is one possible fault model:

$$m = [x, y, z, L, W, \text{strike}, \text{dip}, s_{\text{strike}}, s_{\text{dip}}]$$

Here x, y, z are fault position; L and W are fault length and width; strike and dip are orientation; and s_{strike} and s_{dip} are slip components.

2. Uniform Prior Bounds

setup.py defines a lower and upper bound for every parameter. The prior probability is constant inside the allowed box and zero outside it:

$$p(m) = \text{constant, if } m \text{ is inside all bounds}$$

$$p(m) = 0, \text{ if } m \text{ is outside any bound}$$

$$\log p(m) = 0 \text{ or } -\text{infinity}$$

3. Forward Model

For a proposed model m , the code builds a finite-fault patch. In the GitHub version, it uses `patch_slip_ellip(m)`, which subdivides the fault into small patches and applies an elliptical slip taper.

$$G(m) = \text{predicted InSAR line-of-sight displacement}$$

The elastic dislocation model first predicts 3-D ground displacement, then projects it into the satellite look direction.

4. Residual

Let d be the observed InSAR data and $G(m)$ be the model prediction. The residual is:

$$r = d - G(m)$$

A good model has small residuals.

5. Weighted Gaussian Likelihood

The script scores a model using a weighted quadratic misfit:

$$\log L(m) = -1/2 * r^T S^{-1} B r$$

S^{-1} is the inverse data covariance matrix, based on data uncertainty. B is a dataset-weighting matrix. The smaller the weighted misfit, the higher the likelihood.

6. Posterior Probability

Because the prior is uniform inside the allowed bounds, the posterior is essentially the likelihood inside the box and impossible outside it:

$$\log p(m|d) = -1/2 * (d - G(m))^T S^{-1} B (d - G(m)), \text{ if } m \text{ is allowed}$$

$$\log p(m|d) = -\text{infinity}, \text{ if } m \text{ is outside bounds}$$

7. MCMC Sampling

Quantity	Value in mcmc2.py
Number of walkers	50
Number of steps	50,000
Sampler	emcee ensemble MCMC
Moves	DEMove and DESnookerMove

Each walker proposes new models, computes $\log p(m|d)$, and accepts or rejects proposals. The output is a cloud of samples from the posterior distribution.

8. Final Estimates

After burn-in removal and thinning, the samples are flattened. The script then computes the posterior mean, standard deviation, and 16th, 50th, and 84th percentiles:

$$\text{mean}(m), \text{std}(m), \text{q16}(m), \text{q50}(m), \text{q84}(m)$$

These summarize the best-fitting fault parameters and their uncertainty.

One-Sentence Summary

mcmc2.py repeatedly proposes a fault, predicts its InSAR displacement, compares prediction to data with a weighted misfit, and uses MCMC to map the most probable fault parameters.